

第一次作业第四题原题:

4. Consider a simple linear regression model

$$y_i = \alpha + \beta x_i + \varepsilon_i$$

where $\{x_i\}$ are constants, α and β are parameters, ε_i are iid $N(0, \sigma^2)$ and $i = 1, \dots, n$. A weighted least square estimators for α and β are obtained by minimizing

$$SSE = \sum_{i=1}^n w_i (y_i - \alpha - \beta x_i)^2$$

where $\{w_i\}$ are some predefined known constant values and $\sum_{i=1}^n w_i = 1$. Find the weighted least square estimators for α and β and their variance-covariance matrix.

加上 $w_1, \dots, w_n > 0$ 假设后的求解过程

Solution:

$y_i = \alpha + \beta x_i + \varepsilon_i$ ($i = 1, \dots, n$) 可以写成紧凑形式 $y = \alpha 1_n + \beta x + \varepsilon$, 其中 $\varepsilon \sim N(0_n, \sigma^2 I_n)$

$$\text{记 } \begin{cases} W = \text{diag}\{w_1, \dots, w_n\} \\ w = [w_1, \dots, w_n]^T \\ \gamma = [\alpha, \beta]^T \\ X = [1_n; x] \end{cases} \quad \text{则我们有:}$$

$$y = \alpha 1_n + \beta x + \varepsilon = X\gamma + \varepsilon$$

$$\text{RSS}(\alpha, \beta) := \sum_{i=1}^n w_i (y_i - \alpha - \beta x_i)^2 = (y - X\gamma)^T W (y - X\gamma) \text{ where } W \text{ satisfies } \begin{cases} W \succ 0 \\ \text{tr}(W) = 1 \end{cases}$$

于是 γ 的最小二乘估计量为如下优化问题的全局最小点:

$$\begin{aligned} \hat{\gamma} &:= \arg \min_{\gamma \in \mathbb{R}^2} (y - X\gamma)^T W (y - X\gamma) \\ \nabla_{\gamma} \{(y - X\gamma)^T W (y - X\gamma)\} &= \nabla_{\gamma} \{y^T W y - 2y^T W X \gamma + \gamma^T X^T W X \gamma\} \\ &= -2X^T W y + 2X^T W X \gamma \\ \nabla_{\gamma}^2 \{(y - X\gamma)^T W (y - X\gamma)\} &= 2X^T W X \succ 0 \end{aligned}$$

因此这是一个无约束凸优化问题, 其全局最小点即为驻点.

令 $\nabla_{\gamma} \{(y - X\gamma)^T W (y - X\gamma)\} = -2X^T W y + 2X^T W X \gamma = 0_2$ 可得:

$$\begin{aligned} \hat{\gamma} &= (X^T W X)^{-1} X^T W y \\ &= \begin{bmatrix} 1_n^T W 1_n & 1_n^T W x \\ x^T W 1_n & x^T W x \end{bmatrix}^{-1} \begin{bmatrix} 1_n^T W y \\ x^T W y \end{bmatrix} \quad (\text{note that } \begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \text{ if } ad - bc \neq 0) \\ &= \frac{1}{(1_n^T W 1_n)(x^T W x) - (1_n^T W x)(x^T W 1_n)} \begin{bmatrix} x^T W x & -1_n^T W x \\ -x^T W 1_n & 1_n^T W 1_n \end{bmatrix} \begin{bmatrix} 1_n^T W y \\ x^T W y \end{bmatrix} \quad (\text{note that } \begin{cases} 1_n^T W 1_n = \sum_{i=1}^n w_i = 1 \\ W 1_n = w \end{cases}) \\ &= \frac{1}{x^T W x - (w^T x)^2} \begin{bmatrix} (x^T W x)(w^T y) - (w^T x)(x^T W y) \\ -(w^T x)(w^T y) + x^T W y \end{bmatrix} \end{aligned}$$

因此 α, β 的加权最小二乘估计量 $\hat{\alpha}, \hat{\beta}$ 分别为 $\hat{\gamma}$ 的两个分量:

$$\begin{aligned} \hat{\alpha} &= \frac{(x^T W x)(w^T y) - (w^T x)(x^T W y)}{x^T W x - (w^T x)^2} \\ &= \frac{(\sum_{i=1}^n w_i x_i^2)(\sum_{i=1}^n w_i y_i) - (\sum_{i=1}^n w_i x_i)(\sum_{i=1}^n w_i x_i y_i)}{\sum_{i=1}^n w_i x_i^2 - (\sum_{i=1}^n w_i x_i)^2} \\ \hat{\beta} &= \frac{x^T W y - (w^T x)(w^T y)}{x^T W x - (w^T x)^2} \\ &= \frac{\sum_{i=1}^n w_i x_i y_i - (\sum_{i=1}^n w_i x_i)(\sum_{i=1}^n w_i y_i)}{\sum_{i=1}^n w_i x_i^2 - (\sum_{i=1}^n w_i x_i)^2} \end{aligned}$$

下面我们考虑 $\hat{\gamma} = [\hat{\alpha}, \hat{\beta}]^T$ 的分布:

$$\begin{aligned}\hat{\gamma} &= (X^T W X)^{-1} X^T W y \\ &= (X^T W X)^{-1} X^T W (X\gamma + \varepsilon) \\ &= \gamma + (X^T W X)^{-1} X^T W \varepsilon\end{aligned}$$

$$\begin{aligned}\hat{\gamma} - \gamma &= (X^T W X)^{-1} X^T W \varepsilon \\ &\sim N((X^T W X)^{-1} X^T W \cdot 0_n, (X^T W X)^{-1} X^T W \cdot \sigma^2 I_n \cdot [(X^T W X)^{-1} X^T W]^T) \\ &= N(0_n, \sigma^2 \cdot (X^T W X)^{-1} X^T W^2 X (X^T W X)^{-1})\end{aligned}$$

因此 $\hat{\alpha}, \hat{\beta}$ 的协方差矩阵为:

$$\begin{aligned}\text{Cov}(\hat{\gamma}) &= \sigma^2 \cdot (X^T W X)^{-1} X^T W^2 X (X^T W X)^{-1} \quad (\text{note that } (X^T W X)^{-1} = \frac{1}{x^T W x - (w^T x)^2} \begin{bmatrix} x^T W x & -w^T x \\ -w^T x & 1 \end{bmatrix}) \\ &= \frac{\sigma^2}{[x^T W x - (w^T x)^2]^2} \begin{bmatrix} x^T W x & -w^T x \\ -w^T x & 1 \end{bmatrix} \begin{bmatrix} 1_n^T W^2 1_n & 1_n^T W^2 x \\ x^T W^2 1_n & x^T W^2 x \end{bmatrix} \begin{bmatrix} x^T W x & -w^T x \\ -w^T x & 1 \end{bmatrix}\end{aligned}$$

(不想继续化简了)

一点思考

如果将 $\varepsilon_i \stackrel{iid}{\sim} N(0, \sigma^2)$ ($i = 1, \dots, n$) 的假设变为 " $\varepsilon_i \sim N(0, \frac{\sigma^2}{w_i})$ ($i = 1, \dots, n$) 且它们相互独立",

那么 $\hat{\gamma}$ 的协方差矩阵的形式会更加简单:

(注意此时我们有 $\varepsilon \sim N(0_n, \sigma^2 W^{-1})$)

$$\begin{aligned}\hat{\gamma} &= (X^T W X)^{-1} X^T W y \\ &= (X^T W X)^{-1} X^T W (X\gamma + \varepsilon) \\ &= \gamma + (X^T W X)^{-1} X^T W \varepsilon\end{aligned}$$

$$\begin{aligned}\hat{\gamma} - \gamma &= (X^T W X)^{-1} X^T W \varepsilon \\ &\sim N((X^T W X)^{-1} X^T W \cdot 0_n, (X^T W X)^{-1} X^T W \cdot \sigma^2 W^{-1} \cdot [(X^T W X)^{-1} X^T W]^T) \\ &= N(0_n, \sigma^2 \cdot (X^T W X)^{-1} X^T W X (X^T W X)^{-1}) \\ &= N(0_n, \sigma^2 (X^T W X)^{-1})\end{aligned}$$

我觉得这样的协方差矩阵形式更加简单, 只要取 $W = I_n$ 就是普通最小二乘的情况.

而且这也更符合加权最小二乘用于异方差修正的问题背景.